

SAFETY DATA SHEET

According to OSHA Hazcom Standard 29 CFR 1910.1200

UniCore® 2320

Date of issue: 2021-12-07

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Version: 2.0

1. IDENTIFICATION

A. Product name

- UniCore® 2320

B. Recommended use and restriction on use

- General use : Adhesives, sealants
- Restriction on use : N/A

C. Manufacturer / Supplier / Distributor information

○ Manufacturer information

- Company name : Unitech Co.,Ltd
- Address : 45, Byeolmang-ro 459beon-gil, Danwon-gu ,Ansan-city, Gyeonggi-do, korea
- Emergency telephone number : +82-70-4185-2771

○ Supplier/Distributor information

- Company name : Unitech Co.,Ltd
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2. HAZARD IDENTIFICATION

A. GHS Classification

- Acute toxicity (oral) : Category5
- Acute toxicity (dermal) : Category5
- Acute toxicity (inhalation: vapor) : Category5
- Skin corrosion/irritation : Category2
- Serious eye damage/irritation : Category2
- Skin sensitization : Category1
- Acute aquatic toxicity : Category2
- Chronic aquatic toxicity : Category2

B. GHS label elements

○ Hazard symbols



○ Signal words

- Warning

○ Hazard statements

- H303 May harmful if swallowed.

- H313 May be harmful if contact with skin.
- H315 Causes skin irritation
- H317 May cause an allergic skin reaction
- H319 Causes serious eye irritation
- H333 May be harmful if inhaled.
- H401 Toxic to aquatic organisms.
- H411 Toxic to aquatic life with long lasting effects

○ Precautionary statements

1) Prevention

- P261 Avoid breathing dust/fume/gas/mist/vapours/spray.
- P264 Wash hands thoroughly after handling.
- P272 Contaminated work clothing should not be allowed out of the workplace.
- P273 Avoid release to the environment.
- P280 Wear protective gloves/protective clothing/eye protection/face protection/hearing protection.

2) Response

- P301+P312 IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
- P302+P312 IF ON SKIN: Call a POISON CENTER/doctor if you feel unwell.
- P302+P352 IF ON SKIN: Wash with plenty of soap and water.
- P304+P312 IF INHALED: Call a POISON CENTER or doctor/physician if you feel unwell.
- P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
- P321 Specific treatment
- P332+P313 If skin irritation occurs: Get medical advice/attention.
- P333+P313 If skin irritation or rash occurs: Get medical advice/attention.
- P337+P313 If eye irritation persists: Get medical advice/attention.
- P362+P364 Take off contaminated clothing and wash before reuse.
- P391 Collect spillage.

3) Storage

- Not applicable

4) Disposal

- P501 Dispose of contents/container in accordance with local/regional/national/international regulation

C. Other hazards which do not result in classification

- Not available

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	Trade names and Synonyms	CAS No.	Content(%)
Limestone	Calcium carbonate ; Calcite ; Marble	1317-65-3	30 ~ 40
4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane	Poly(bisphenol-A-co-epichlorohydrin)	25068-38-6	30 ~ 40
2-Propenenitrile polymer with 1,3-butadiene, carboxy-terminated polymer with bisphenol A and epichlorohydrin	Bisphenol A, epichlorohydrin polymer, reaction product with carboxyl-terminated butadiene, acrylonitrile polymer	68610-41-3	7 ~ 10
Cyanoguanidine	N-Cyanoguanidine ; Dicyanodiamide ; 1-Cyanoguanidine ; 2-Cyanoguanidine ; Dicyanediamide	461-58-5	5 ~ 8
Calcium oxide	Quick lime ; Lime ; Burnt lime ; Unslaked lime ; Calcium monoxide	1305-78-8	1 ~ 2

8,18-Dichloro-5,15-diethyl-5,15-dihydrodiindolo[3,2-b:3',2'-m]triphenodioxazine (C.I. pigment violet	Diindolo[3,2-b:3',2'-m]triphenodioxazine, 8,18-dichloro-5,15-diethyl-5,15-dihydro- ; 8,18-Dichloro-5,15-diethyl-5,15-dihydrodiindolo[3,2-b:3',2'-m]triphenodioxazine ; C.I. PIGMENT VIOLET 23,Diindolo[3,2-b:3',2'-m]triphenodioxazine, 8,18-dichloro-5,15-diethyl-5,15-dihydro- ; Carbazole violet ; Dioxazine violet ;	6358-30-1	0.01 ~ 0.05
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4. FIRST AID MEASURES

A. Eye contact

- Do not rub your eyes.
- Immediately flush eyes with plenty of water for at least 15 minutes and call a doctor/physician.
- Go to the hospital immediately if symptoms(flare, irritate) occur.
- Remove contact lenses if worn.

B. Skin contact

- Flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes.
- Wash contaminated clothing thoroughly before re-using.
- Go to the hospital immediately if symptoms(flare, irritate) occur.
- Wash thoroughly after handling.

C. Inhalation contact

- Take specific treatment if needed.
- When exposed to large amounts of steam and mist, move to fresh air.

D. Ingestion contact

- Please be advised by doctor whether induction of vomit is demanded or not.
- Rinse your mouth with water immediately.

E. Delayed and immediate effects and also chronic effects from short and long term exposure

- Not available

F. Notes to physician

- Notify medical personnel of contaminated situations and have them take appropriate protective measures.

5. FIREFIGHTING MEASURES

A. Suitable (Unsuitable) extinguishing media

- Avoid use of water jet for extinguishing
- Dry chemical, carbon dioxide, regular foam extinguishing agent, spray

B. Specific hazards arising from the chemical

- Causes serious eye irritation
- Causes skin irritation
- May be harmful if contact with skin.
- May be harmful if inhaled.
- May cause an allergic skin reaction

C. Special protective actions for firefighters

- Avoid inhalation of materials or combustion by-products.
- Cool containers with water until well after fire is out.
- Do not approach the tank surrounded by fire until it is extinguished.
- In case of conflagration, use automatic fire sprinkler. Major fire may require withdrawal, allowing the object itself to burn.
- Keep unauthorized personnel out.

6. ACCIDENTAL RELEASE MEASURES

A. Personal precautions, protective equipment and emergency procedures

- Do not touch spilled material. Stop leak if you can do it without risk.
- Handle the damaged containers or spilled material after wearing appropriate protective equipment
- Move container to safe area from the leak area.
- Must work against the wind, let the upwind people to evacuate.
- Remove all sources of ignition.

B. Environmental precautions

- If large amounts have been spilled, inform the relevant authorities.
- Prevent runoff and contact with waterways, drains or sewers.

C. Methods and materials for containment and cleaning up

- Appropriate container for disposal of spilled material collected.
- Dike for later disposal.
- Disposal of waste shall be in compliance with the Wastes Control Act
- Large spill : Stay upwind and keep out of low areas. Dike for later disposal.
- Notify the central and local government if the emission reach the standard threshold.

7. HANDLING AND STORAGE

A. Precautions for safe handling

- Avoid contact with incompatible materials.
- Avoid direct physical contact.
- Comply with all applicable laws and regulations for handling
- Dealing only with a well-ventilated place.
- Do not handle until all safety precautions have been read and understood.

B. Conditions for safe storage, including any incompatibilities

- Avoid direct sunlight.
- Check regularly for leaks.
- Do not apply any physical shock to container.
- Do not apply direct heat.
- Do not use damaged containers.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

A. Exposure limits

- ACGIH TLV
 - [Calcium oxide] : TWA, 2 mg/m³
- OSHA PEL
 - [Limestone] : 15 mg/m³ (Total dust), 5 mg/m³ (Respirable fraction)
 - [Calcium oxide] : 5 mg/m³

B. Engineering controls

- Business owner is recommended to maintain below recommended exposure limits for the working place with general exhaust of gas/vapour/mist/fume.

C. Individual protection measures, such as personal protective equipment

- Respiratory protection
 - Any air-purifying respirator with a full facepiece and an organic vapor canister.
 - Any chemical cartridge respirator with a full facepiece and organic vapor cartridge(s).
 - Any chemical cartridge respirator with organic vapor cartridge(s).
 - Consider warning properties before use.
 - For Unknown Concentration or Immediately Dangerous to Life or Health : Any supplied-air respirator with full facepiece and operated in a pressure-demand or other positive-pressure mode in combination with a separate escape supply. Any self-contained breathing apparatus with a full facepiece.
 - Respiratory protection is ranked in order from minimum to maximum.

- Under conditions of frequent use or heavy exposure, Respiratory protection may be needed.
- Eye protection
 - Provide an emergency eye wash station and quick drench shower in the immediate work area.
 - Wear primary eye protection such as splash resistant safety goggles with a secondary protection face shield.
- Hand protection
 - Wear appropriate chemical resistant glove.
- Skin protection
 - Wear appropriate chemical resistant protective clothing.
- Others
 - Not available

9. PHYSICAL AND CHEMICAL PROPERTIES

A. Appearance	
- Appearance	paste
- Color	Violet
B. Odor	Not available
C. Odor threshold	Not available
D. pH	Not available
E. Melting point/Freezing point	Not available
F. Initial Boiling Point/Boiling Ranges	Not available
G. Flash point	Not available
H. Evaporation rate	Not available
I. Flammability(solid, gas)	Not available
J. Upper/Lower Flammability or explosive limits	Not available
K. Vapour pressure	Not available
L. Solubility	Not available
M. Vapour density	Not available
N. Specific gravity(Relative density)	In less than 1.3
O. Partition coefficient of n-octanol/water	Not available
P. Autoignition temperature	Not available
Q. Decomposition temperature	Not available
R. Viscosity	340 - 400만 cps
S. Molecular weight	Not available

10. STABILITY AND REACTIVITY

A. Chemical Stability

- This material is stable under recommended storage and handling conditions.

B. Possibility of hazardous reactions

- Hazardous Polymerization will not occur.

C. Conditions to avoid

- Avoid : Accumulation of electrostatic charges, Heating, Flames and hot surfaces
- Avoid contact with incompatible materials and condition.

D. Incompatible materials

- Not available

E. Hazardous decomposition products

- May emit flammable vapour if involved in fire.

11. TOXICOLOGICAL INFORMATION

A. Information on the likely routes of exposure

- ☐ Respiratory tracts
 - Not available
- ☐ Oral
 - May harmful if swallowed.
- ☐ Eye-Skin
 - Causes skin irritation
 - May cause an allergic skin reaction

B. Delayed and immediate effects and also chronic effects from short and long term exposure

- ☐ Acute toxicity
 - * Oral
 - Product (ATEmix) : 2000mg/kg < ATEmix <= 5000mg/kg
 - [Limestone] : LD50 > 2000 mg/kg Rat No death Not classified (OECD TG 420, GLP) (ECHA)
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : LD50 >2000 mg/kg Rat (OECD Guideline 420)(ECHA)
 - [Cyanoguanidine] : LD50 > 30000 mg/kg Rat (SIDS, IUCLID)
 - [Calcium oxide] : LD50 > 2000 mg/kg Rat (OECD TG 425, GLP) (ECHA)
 - * Dermal
 - Product (ATEmix) : 2000mg/kg < ATEmix <= 5000mg/kg
 - [Limestone] : LD50 > 2000 mg/kg Rat No death Not classified (OECD TG 402, GLP) (ECHA)
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : LD50 >2000 mg/kg Rat (CHEMIDPLUS)
 - [Cyanoguanidine] : LD50 > 10000 mg/kg Rabbit (KOSHA)
 - [Calcium oxide] : LD50 > 2500 mg/kg Guinea pig (OECD TG 402) (ECHA)
 - * Inhalation
 - Product (ATEmix) : 20.0mg/L < ATEmix <= 50.0mg/L, Vapour, 4hr
 - [Limestone] : Aerosol LC50 > 3 mg/L air 4 hr Rat No death Not classified (OECD TG 403, GLP) (ECHA)
 - [Calcium oxide] : Dust LC50 > 6.04 mg/L 4 hr Rat (Cat. 5) (ECHA)
- ☐ Skin corrosion/irritation
 - Causes skin irritation
- ☐ Serious eye damage/irritation
 - Causes serious eye irritation
- ☐ Respiratory sensitization
 - Not available
- ☐ Skin sensitization
 - May cause an allergic skin reaction
- ☐ Carcinogenicity
 - * IARC
 - Not available
 - * OSHA
 - Not available
 - * ACGIH
 - Not available
 - * NTP
 - Not available
 - * EU CLP

- Not available
- ☐ Germ cell mutagenicity
 - Not available
- ☐ Reproductive toxicity
 - Not available
- ☐ STOT-single exposure
 - Not available
- ☐ STOT-repeated exposure
 - Not available
- ☐ Aspiration hazard
 - Not available

12. ECOLOGICAL INFORMATION

A. Ecotoxicity

- ☐ Fish
 - [Cyanoguanidine] : LC50 >100 mg/l 96 hr *Oryzias latipes* (SIDS)
- ☐ Crustaceans
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : EC50 ca. 2 mg/l 48 hr *Daphnia magna* (OECD TG 202) (ECHA)
 - [Cyanoguanidine] : EC50 >1000 mg/l 48 hr *Daphnia magna*(OECD TG 202) (OECD SIDS)
- ☐ Algae
 - [Limestone] : EC50 > 14 mg/L, NOEC 14 mg/L 72 hr *Desmodesmus subspicatus* (OECD TG 201, GLP) (ECHA)
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : EC50 ca. 9 mg/L 48 hr *Scenedesmus capricornutum* (ECHA)
 - [Cyanoguanidine] : EC50 935 mg/l 72 hr *Selenastrum capricornutum*(OECD TG 201) (OECD SIDS)
 - [Calcium oxide] : EC50 184.57 mg/L 72 hr *Raphidocelis subcapitata* (EU Method C.3, GLP) (Read-across 1305-62-0) (ECHA)

B. Persistence and degradability

- ☐ Persistence
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : log Kow 3 (25°C) (ECHA)
 - [Cyanoguanidine] : log Kow = -1.15 (NLM/HSDB)
- ☐ Degradability
 - Not available

C. Bioaccumulative potential

- ☐ Bioaccumulative potential
 - [Cyanoguanidine] : BCF ≤3.1 (*Cyprinus carpio*)(IUCLID, OECD Screening Information Data Set)
- ☐ Biodegradation
 - [Limestone] : 90 % 28 d (CO2 evolution) Readily biodegradable (ECHA)
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : ca.5 % 28day (O2 consumption) Not Readily biodegradable (ECHA)
 - [Cyanoguanidine] : Biodegradability = 0 (%) 28 day (OECD Screening Information Data Set)

D. Mobility in soil

- Not available

E. Other adverse effects

- Not available

13. DISPOSAL CONSIDERATIONS

A. Disposal methods

- It shall be treated by incineration
- Oil water separation technology shall be applied as pre-waste treatment if it is applicable

- Stabilization and minimization treatment by incineration or similar method can be applied, if more than two kinds of designated wastes are in mixture state and it is impractical to separate them

B. Special precautions for disposal

- Anyone with business license number who generates industrial wastes shall treat the waste by him/herself or by entrusting to the legal entities who treat the wastes, recycle the wastes of others or install and operate the waste treatment facilities according to the Wastes Control Act
- Dispose of waste in accordance with all applicable laws and regulations.

14. TRANSPORT INFORMATION

A. UN No. (IMDG CODE/IATA DGR)

- 3082

B. Proper shipping name

- ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

C. Hazard Class

- 9

D. IMDG CODE/IATA DGR Packing group

- III

E. Marine pollutant

- Applicable

F. Special precautions for user related to transport or transportation measures

- Local transport follows in accordance with Dangerous goods Safety Management Law.
- Package and transport follow in accordance with Department of Transportation (DOT) and other regulatory agency requirements.
- EmS FIRE SCHEDULE : F-A (General fire schedule)
- EmS SPILLAGE SCHEDULE : S-F (Water-soluble marine pollutants)

15. REGULATORY INFORMATION

A. National and/or international regulatory information

- POPs Management Law
 - [Limestone] : Not applicable
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : Not applicable
 - [2-Propenenitrile polymer with 1,3-butadiene, carboxy-terminated polymer with bisphenol A and epichlorohydrin] : Not applicable
 - [Cyanoguanidine] : Not applicable
 - [Calcium oxide] : Not applicable
 - [8,18-Dichloro-5,15-diethyl-5,15-dihydrodiindolo[3,2-b:3',2'-m]triphenodioxazine (C.I. pigment violet 023)] : Not applicable
- Information of EU Classification
 - * Classification
 - [4,4'-(1-methylethylidene)bisphenol polymer with (chloromethyl)oxirane] : H315,H317,H319,H411
- U.S. Federal regulations
 - * OSHA PROCESS SAFETY (29CFR1910.119)
 - Not applicable
 - * CERCLA Section 103 (40CFR302.4)
 - Not applicable
 - * EPCRA Section 302 (40CFR355.30)
 - Not applicable
 - * EPCRA Section 304 (40CFR355.40)
 - Not applicable
 - * EPCRA Section 313 (40CFR372.65)

- Not applicable
- ☐ Rotterdam Convention listed ingredients
 - Not applicable
- ☐ Stockholm Convention listed ingredients
 - Not applicable
- ☐ Montreal Protocol listed ingredients
 - Not applicable

16. OTHER INFORMATION

A. Reference

- The information contained herein is believed to be accurate. It is provided independently of any sale of the product for purpose of hazard communication. It is not intended to constitute performance information concerning the product. No express warranty, or implied warranty of merchantability or fitness for a particular purpose is made with respect to the product or the information contained herein.
- This Safety Data Sheet was compiled with data and information from the following sources: KOSHA, NITE, ESIS, NLM, SIDS, IPCS

B. Issue date

- 2021-12-07

C. Revision number and Last date revised

- 1 times, 2024-04-26

D. Other

- This SDS is prepared according to the Globally Harmonized System (GHS).